

1. Very often the surface of the mild steel is hardened by keeping the steel surface in contact with carbon rich material (from where carbon diffuses into steel) at very high temperature in a furnace. Figure (i) below shows the schematic of such process, where b is the width, L is the length of mild steel rod of uniform thickness (rod-length L is very large as compared to other dimension of the rod). The initial concentration of carbon in steel rod is w_i (in mass %) = 0.15%. Assume that the interface at $x=0$ always remains at equilibrium such that the steel-side carbon local concentration always remain at, w_s (in mass %) = 1.2%.

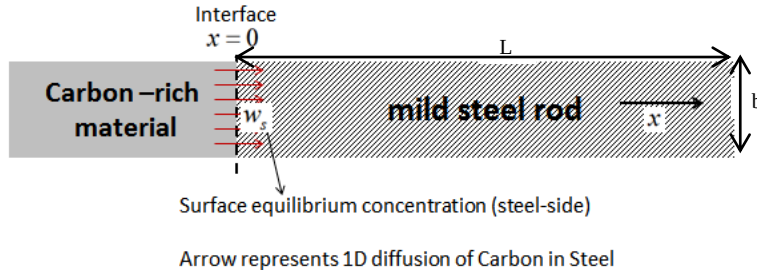


Figure (i)

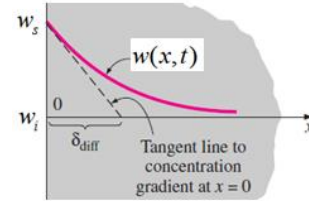


Figure (ii)

Assuming one dimensional mass-diffusion in the X -direction, answer the following for very early times ($t \ll L^2/D_{cs}$):

- (a) Write the governing equation and boundary condition. (consider $w(x, t)$ = is the local concentration (mass %) any location x and time t).
- (b) Solve the governing equation and prove that the penetration depth (δ_{diff}) is given by $\delta_{diff} = (\pi D_{cs} t)^{0.5}$, where t is the real time. Penetration depth is defined as the location x , where the tangent to concentration profile ($w(x, t)$) for any t at $x=0$ intersects the line $w = w_i$ as shown in figure (ii), above.
- (c) How long it will take to reach w (in mass %) = 1% at $x = 0.5\text{mm}$ from the interface in steel side. The other details are given below: Diffusivity of Carbon in mild-steel, $D_{cs} = 4.7 \times 10^{-10} \text{ m}^2/\text{s}$.

Error Function/complementary Error function is given by :

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-t^2) dt, \quad \text{erfc}(z) = 1 - \text{erf}(z). \quad \frac{d(\text{erf}(z))}{dz} = \frac{2}{\sqrt{\pi}} \exp(-z^2)$$

The complementary error function

ξ	$\text{erfc}(\xi)$	ξ	$\text{erfc}(\xi)$	ξ	$\text{erfc}(\xi)$	ξ	$\text{erfc}(\xi)$	ξ	$\text{erfc}(\xi)$	ξ	$\text{erfc}(\xi)$
0.00	1.00000	0.38	0.5910	0.76	0.2825	1.14	0.1069	1.52	0.03159	1.90	0.00721
0.02	0.9774	0.40	0.5716	0.78	0.2700	1.16	0.10090	1.54	0.02941	1.92	0.00662
0.04	0.9549	0.42	0.5525	0.80	0.2579	1.18	0.09516	1.56	0.02737	1.94	0.00608
0.06	0.9324	0.44	0.5338	0.82	0.2462	1.20	0.08969	1.58	0.02545	1.96	0.00557
0.08	0.9099	0.46	0.5153	0.84	0.2349	1.22	0.08447	1.60	0.02365	1.98	0.00511
0.10	0.8875	0.48	0.4973	0.86	0.2239	1.24	0.07950	1.62	0.02196	2.00	0.00468
0.12	0.8652	0.50	0.4795	0.88	0.2133	1.26	0.07476	1.64	0.02038	2.10	0.00298
0.14	0.8431	0.52	0.4621	0.90	0.2031	1.28	0.07027	1.66	0.01890	2.20	0.00186
0.16	0.8210	0.54	0.4451	0.92	0.1932	1.30	0.06599	1.68	0.01751	2.30	0.00114
0.18	0.7991	0.56	0.4284	0.94	0.1837	1.32	0.06194	1.70	0.01612	2.40	0.00069
0.20	0.7773	0.58	0.4121	0.96	0.1746	1.34	0.05809	1.72	0.01500	2.50	0.00041
0.22	0.7557	0.60	0.3961	0.98	0.1658	1.36	0.05444	1.74	0.01387	2.60	0.00024
0.24	0.7343	0.62	0.3806	1.00	0.1573	1.38	0.05098	1.76	0.01281	2.70	0.00013
0.26	0.7131	0.64	0.3654	1.02	0.1492	1.40	0.04772	1.78	0.01183	2.80	0.00008
0.28	0.6921	0.66	0.3506	1.04	0.1413	1.42	0.04462	1.80	0.01091	2.90	0.00004
0.30	0.6714	0.68	0.3362	1.06	0.1339	1.44	0.04170	1.82	0.01006	3.00	0.00002
0.32	0.6509	0.70	0.3222	1.08	0.1267	1.46	0.03895	1.84	0.00926	3.20	0.00001
0.34	0.6306	0.72	0.3086	1.10	0.1198	1.48	0.03635	1.86	0.00853	3.40	0.00000
0.36	0.6107	0.74	0.2953	1.12	0.1132	1.50	0.03390	1.88	0.00784	3.60	0.00000

$\text{erf}(\infty)=1$.